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CANDIDATE NAME:	Praise Mutema
SCHOOL/INSTITUTION:	Mopani park secondary school
EXAM BOARD:	ZIMSEC
SUBJECT:	MATHEMATICS
LEVEL / GRADE:	O-LEVEL
PAPER:	Paper 1 — Non-Calculator (Short Answer & Structured)
TOTAL MARKS:	100 marks
TIME ALLOWED:	2h 30min

ZIMSEC Mathematics O-Level — Paper 1 Mock Examination

Topic: Concentration

INSTRUCTIONS TO CANDIDATES

1. Write your name and school in the spaces provided on the front page.
2. This paper has THREE sections. Answer ALL questions in every section.
3. Total marks available: 100. Time allowed: 2h 30min.
4. Show ALL working for full marks — correct answers without working may not receive full credit.
5. Write clearly and legibly. Crossed-out work will not be marked.
6. If a question has sub-parts (a), (b), (c), answer ALL sub-parts.
7. Use correct units and significant figures where applicable.

ZIMSEC O-Level Mathematics Paper 1 (Non-Calculator) Examination

Topic: Concentration

Total marks: 100 | Time: 2h 30min

SECTION A — Short Answer [5 questions × 3 marks = 15 marks]

1. Q1. A container holds 1200 liters of water. Water flows in at a rate of 2 liters per minute and flows out at a rate of 1.5 liters per minute. Calculate the concentration of water in the container after 60 minutes.

2. Q2. A tank has a capacity of 2000 liters. It is initially filled with 1500 liters of water. Water flows in at a rate of $(\frac{2}{3})$ liters per minute. Calculate the concentration of water in the tank after 45 minutes.

3. Q3. A bucket contains 500 grams of sugar. 20 grams of sugar are added every minute for 15 minutes. Calculate the concentration of sugar in the bucket after 15 minutes.

4. Q4. A solution contains 60 grams of salt in 800 grams of water. How many grams of salt are in 1 liter of the solution?

5. Q5. A container holds 3000 liters of a solution containing $(\frac{1}{5})$ grams of salt per liter. How many grams of salt are in 1000 liters of the solution?

SECTION B — Structured Questions [4 questions × 6 marks = 24 marks]

1. Q6. A swimming pool has a capacity of 12000 liters. Water flows in at a rate of 3 liters per minute and flows out at a rate of $(\frac{2}{3})$ liters per minute. Calculate the concentration of water in the pool after 60 minutes.

(a) Calculate the amount of water in the pool after 60 minutes. (2 marks)

(b) Calculate the concentration of water in the pool after 60 minutes. (2 marks)

(c) Calculate the volume of water added to the pool in 60 minutes. (2 marks)

2. Q7. A tank has a capacity of 1500 liters. It is initially filled with 1000 liters of water. Water flows in at a rate of $(\frac{3}{4})$ liters per minute. Calculate the concentration of water in the tank after 45 minutes.

(a) Calculate the amount of water in the tank after 45 minutes. (2 marks)

(b) Calculate the concentration of water in the tank after 45 minutes. (2 marks)

(c) Calculate the volume of water added to the tank in 45 minutes. (2 marks)

SECTION C — Extended Response [1 question × 61 marks = 61 marks]

1. Q8. A container holds 4000 liters of a solution containing $(\frac{1}{8})$ grams of salt per liter. Water flows in at a rate of 2 liters per minute and flows out at a rate of $(\frac{3}{5})$ liters per minute. Calculate the concentration of salt in the solution after 120 minutes. Justify your answer by explaining the process of calculating the concentration.

MARKING SCHEME

This marking scheme is intended to assist teachers and students in self-assessment. Answers may vary — credit any valid equivalent response.

MARKING SCHEME FOR ZIMSEC O-LEVEL MATHEMATICS PAPER 1 (NON-CALCULATOR) EXAMINATION

SECTION A - SHORT ANSWER

1. Q1. A container holds 1200 liters of water. Water flows in at a rate of 2 liters per minute and flows out at a rate of 1.5 liters per minute. Calculate the concentration of water in the container after 60 minutes.

Answer:

- Let x be the amount of water in the container after 60 minutes.
- Water flows in at a rate of 2 liters per minute, so in 60 minutes, $2 \times 60 = 120$ liters of water flow in.
- Water flows out at a rate of 1.5 liters per minute, so in 60 minutes, $1.5 \times 60 = 90$ liters of water flow out.
- The net change in the amount of water in the container is $120 - 90 = 30$ liters.
- Therefore, the amount of water in the container after 60 minutes is $1200 + 30 = 1230$ liters.
- The concentration of water in the container after 60 minutes is $1230 / 1200 = 1.025$ or 102.5%.

Marks: 3

2. Q2. A tank has a capacity of 2000 liters. It is initially filled with 1500 liters of water. Water flows in at a rate of $(2/3)$ liters per minute. Calculate the concentration of water in the tank after 45 minutes.

Answer:

- Let x be the amount of water in the tank after 45 minutes.
- Water flows in at a rate of $(2/3)$ liters per minute, so in 45 minutes, $(2/3) \times 45 = 30$ liters of water flow in.
- The amount of water in the tank after 45 minutes is $1500 + 30 = 1530$ liters.
- The concentration of water in the tank after 45 minutes is $1530 / 2000 = 0.765$ or 76.5%.

Marks: 3

3. Q3. A bucket contains 500 grams of sugar. 20 grams of sugar are added every minute for 15 minutes. Calculate the concentration of sugar in the bucket after 15 minutes.

Answer:

- Let x be the amount of sugar in the bucket after 15 minutes.
- 20 grams of sugar are added every minute for 15 minutes, so $20 \times 15 = 300$ grams of sugar are added.
- The amount of sugar in the bucket after 15 minutes is $500 + 300 = 800$ grams.
- The concentration of sugar in the bucket after 15 minutes is $800 / 500 = 1.6$ or 160%.

Marks: 3

4. Q4. A solution contains 60 grams of salt in 800 grams of water. How many grams of salt are in 1 liter of the solution?

Answer:

- The density of the solution is $(60 + 800) / 800 = 860 / 800 = 1.075$ grams per milliliter (mL).
- Since 1 liter is equal to 1000 mL, the amount of salt in 1 liter of the solution is $1.075 \times 1000 = 1075$ grams.

Marks: 3

5. Q5. A container holds 3000 liters of a solution containing $(1/5)$ grams of salt per liter. How many grams of salt are in 1000 liters of the solution?

Answer:

- The amount of salt in 1000 liters of the solution is $(1/5) \times 1000 = 200$ grams.

Marks: 3

SECTION B - STRUCTURED QUESTIONS

1. Q6. A swimming pool has a capacity of 12000 liters. Water flows in at a rate of 0.5 liters per minute and flows out at a rate of 0.2 liters per minute. Calculate the concentration of water in the pool after 90 minutes.

Answer:

- Let x be the amount of water in the pool after 90 minutes.
- Water flows in at a rate of 0.5 liters per minute, so in 90 minutes, $0.5 \times 90 = 45$ liters of water flow in.
- Water flows out at a rate of 0.2 liters per minute, so in 90 minutes, $0.2 \times 90 = 18$ liters of water flow out.
- The net change in the amount of water in the pool is $45 - 18 = 27$ liters.
- Therefore, the amount of water in the pool after 90 minutes is $12000 + 27 = 12027$ liters.
- The concentration of water in the pool after 90 minutes is $12027 / 12000 = 1.002125$ or 100.2125%.

Marks: 6

2. Q7. A tank has a capacity of 5000 liters. It is initially filled with 3000 liters of water. Water flows in at a rate of $(1/4)$ liters per minute. Calculate the concentration of water in the tank after 60 minutes.

Answer:

- Let x be the amount of water in the tank after 60 minutes.
- Water flows in at a rate of $(1/4)$ liters per minute, so in 60 minutes, $(1/4) \times 60 = 15$ liters of water flow in.
- The amount of water in the tank after 60 minutes is $3000 + 15 = 3015$ liters.
- The concentration of water in the tank after 60 minutes is $3015 / 5000 = 0.603$ or 60.3%.

Marks: 6

3. Q8. A bucket contains 2000 grams of sugar. 15 grams of sugar are added every 5 minutes. Calculate the concentration of sugar in the bucket after 60 minutes.

Answer:

- Let x be the amount of sugar in the bucket after 60 minutes.
- 15 grams of sugar are added every 5 minutes, so in 60 minutes, $15 \times (60/5) = 180$ grams of sugar are added.
- The amount of sugar in the bucket after 60 minutes is $2000 + 180 = 2180$ grams.
- The concentration of sugar in the bucket after 60 minutes is $2180 / 2000 = 1.09$ or 109%.

Marks: 6

4. Q9. A solution contains 120 grams of salt in 900 grams of water. How many grams of salt are in 2 liters of the solution?

Answer:

- The density of the solution is $(120 + 900) / 900 = 1020 / 900 = 1.1333$ grams per milliliter (mL).
- Since 1 liter is equal to 1000 mL, the amount of salt in 2 liters of the solution is $1.1333 \times 2000 = 2266.67$ grams.

Marks: 6

TOTAL MARKS: 100